

Consumers' adoption of AR-based virtual fitting rooms: from the perspective of theory of interactive media effects

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adoption of
AR-based
VFRs

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Abstract

Purpose – The purpose of this study was to investigate the relationship between consumers' perceived media characteristics, telepresence, attitudes and adoption intention towards augmented reality (AR)-based virtual fitting rooms (VFRs). Additionally, the mediating effect of telepresence was examined.

Design/methodology/approach – Data were collected from 352 university students and analysed using structural equation modelling.

Findings – Empirical results suggest significant positive influences of media characteristics, including perceived interactivity and augmentation, on telepresence, which, in turn, influenced attitudes and adoption intention towards AR-based VFRs. Also, telepresence mediated the relationship between media characteristics and consumers' attitudes.

Research limitations/implications – Data for this study were collected based on the subjects' one-time experience with a particular AR-based VFR. Therefore, the generalisation of the findings may be limited.

Practical implications – An important implication is that the enhancement of rendering interactive and augmented features is crucial for adoption of AR-based VFRs considering the key role of interactivity and augmentation in inducing telepresence, attitudes and adoption intention.

Originality/value – The paper empirically tested the importance of unique media characteristics, telepresence and attitudes in consumers' adoption of AR-based VFRs through the lens of the theory of interactive media effects.

Keywords Telepresence, Interactivity, Attitudes, Adoption, Virtual fitting room, Augmentation

Paper type Research paper

Introduction

Virtual fitting rooms (VFRs) are among the most promising technologies for overcoming the inherent problems of online shopping – the inability to try on an item before making a purchase (Blazquez, 2014; Kim and Forsythe, 2008). Using VFRs, consumers can virtually try on products using a personalised model, which is created and/or simulated based on their own body measurements through various data input methods (Gültepe and Gündükbay, 2014; Kim and Forsythe, 2008; Lee and Xu, 2019). Among several different types of VFRs available in the marketplace, the new trend in augmented reality (AR) has recently led to the booming of AR-based VFRs (Lee and Xu, 2019; Pachoulakis and Kapetanaki, 2012). AR-based VFRs can provide a “mixed reality” through capturing real-world data of users and their surrounding environment by using a camera-based technology embedded in a device, whereas the objects portrayed in the environment are virtually simulated and superimposed onto the user's augmented body (Bonetti *et al.*, 2018; Cho and Schwarz, 2010; Yim *et al.*, 2017).

Although AR-based VFRs have the potential to significantly affect consumers' online shopping experiences with enhanced interactive features and the ability to better immerse



consumers (Bonetti *et al.*, 2018; Yim *et al.*, 2017), they are still in the early stages of consumers adoption and are yet to be widely implemented in the fashion industry. Moreover, research is limited in examining and explaining the psychological mechanism behind consumers' adoption process of VFRs. Previous studies largely focused on technical and functional aspects of VFRs in general, investigating methods of improving the congruity between visualised models and consumers (Gültepe and Güdükbay, 2014). Moreover, most researchers studied VFRs that virtually visualise the entire setting, including users, products and environments, characterised by 3D virtual simulation, limiting the scope of their study to VFRs that primarily utilise interactive 3D virtual simulation techniques (Lee and Park, 2017; Merle *et al.*, 2012).

With unique characteristics of AR-based VFRs that combine AR features and 3D interactive features, consumers can follow different psychological evaluative process (Javornik, 2016; Lee and Xu, 2019). However, attention was paid to individual media characteristics of technologies one at a time, either interactivity (e.g. Fiore and Jin, 2003) or augmentation (e.g. Javornik, 2016). Considering the technically mixed nature of AR-based VFRs, it is critical to understand how combined media characteristics, both interactivity and augmentation, contribute to consumers' experience and elicit consumer adoption intention towards VFRs (Javornik, 2016). With this in mind, the media-effect approach, the theory of interactive media effects (TIME) (Sundar *et al.*, 2015) was adopted in this study to investigate the influence of both interactivity and augmentation on fashion consumers' adoption towards AR-based VFRs.

According to TIME, media characteristics can serve as the catalyst for consumers to elicit psychological correlates, to be immersed into the media, and consequently to show cognitive, affective and conative responses in turn (Javornik, 2016; Sundar *et al.*, 2015). In other words, psychological correlates, derived from unique media characteristics or features, are predictors that explain the impact of technology on consumers as well as the interactions between the technology and consumers (Javornik, 2016). In addition, TIME proposes that the key mechanism that enables consumers to elicit cognitive, affective and conative responses towards a technology is consumers' immersion when using the technology, explaining consumers' immersive experience as an intermediate variable. In the VFR literature, telepresence, "the extent to which one feels present in the mediated environment" (Steuer, 1992, pp.76), has often been used to measure consumers' immersive experience and was found to play a mediating role between media characteristics such as image interactivity and consumer attitude (Fiore *et al.*, 2005; Li *et al.*, 2002; Suh and Chang, 2006). Telepresence can have a positive impact on consumers' attitudes towards a product or service as consumers can perceive that the artificial environment provides necessary cognitive and sensory input equivalent to that of the real environment (Fiore *et al.*, 2005). Thus, telepresence can lead consumers to perceive instrumental and experiential value, which can influence their attitudes towards a product or service in turn (Mollen and Willson, 2009). As VFRs have the capability to offer both utilitarian and hedonic benefits with instrumental and experiential features (Kim and Forsythe, 2008; Lee *et al.*, 2010), both the utilitarian and hedonic components of consumer's attitudes can be influenced by the way consumers perceive media characteristics and the extent to which they feel telepresence. In addition, given that media characteristics are regarded as key factors affecting the formation of immersive experience, and that immersive experience is a significant driver of consumers' cognitive, affective and behavioural responses, more detailed examination of these relationships are necessary.

Therefore, the purpose of this study was to explain the process of consumers' adoption of AR-based VFRs for online shopping through the lens of TIME. Specifically, this study aimed to examine (a) the influences of consumers' perceptions of media characteristics of AR-based VFRs on telepresence, (b) the influences of telepresence on both utilitarian and hedonic attitudes and (c) the consequent influences of consumers' attitudes on the adoption intention

towards AR-based VFRs. Additionally, this study further investigated the mediating effect of telepresence on consumers' adoption process.

Background and theoretical framework

Emerging AR-based virtual fitting rooms (VFRs) in the fashion industry

Today, online shopping is increasingly gaining popularity with the exponential growth of e-commerce (Statista, 2019). However, online shopping bears significant risk because consumers cannot try-on products before purchase (Lee and Xu, 2019). Online shopping poses challenges for fashion consumers mainly in terms of style and fit of the products, knowledge of which is typically acquired through direct interaction with products (Fiore and Jin, 2003). Aiming to overcome this challenge, a growing number of online retailers have started to use VFRs to provide consumers with a virtual try-on experience during their online shopping. A variety of global fashion brands and retailers, such as Land's End, Uniqlo, Rebecca Minkoff and Timberland, provide VFRs in their online stores to meet consumers' unmet fitting demand and to showcase VFRs in their key brick-and-mortar stores (Caboni and Hagberg, 2019; Lee and Xu, 2019).

First introduced in 2005, VFRs were not widely reported or studied until 2010 with the advancement of technologies and the growth of e-commerce (Alfredo and Rodriguez, 2016; Bonetti et al., 2018). Until the late 2000s, VFRs were primarily image interactive technologies (IIT), which primarily utilise 3D virtual simulation techniques, characterised by 3D product visualisation, rotation properties and mix-and-match ability (i.e. Fiore and Jin, 2003; Lee and Park, 2017; Merle et al., 2012; Park et al., 2008). Using 3D virtual simulation, which is employed to create virtual reality (VR), IIT primarily focused on virtually visualising the entire setting, including users, products and environment by placing users into a simulated 3D world through creating virtual models in the computer-mediated environment (Bonetti et al., 2018; Lee and Leonas, 2018; Lee et al., 2010; Merle et al., 2012). Recently, a more advanced type of VFR evolved to incorporate AR with the features of 3D visual technologies to offer more realistic experiences (Caboni and Hagberg, 2019; Pachoulakis and Kapetanaki, 2012). Global fashion retailers, including Rebecca Minkoff and Zara, are testing AR-based VFRs, offering mobile-based applications (Apps) for online shopping (Steele, 2019). According to Faust et al. (2012, pp. 1164), AR is characterised by "the superposition of virtual objects (e.g. computer-generated images) on the real environment of the user". Unlike VR-based VFRs, AR-based VFRs can provide a "mixed reality" through capturing real-world data of users and their surrounding environment by using a camera-based technology embedded in a device, whereas the objects portrayed in the environment are virtually simulated and superimposed onto the user's augmented body (Bonetti et al., 2018; Cho and Schwarz, 2010; Yim et al., 2017). Thus, AR can be regarded as an extension of VR (Chen et al., 2011).

Similarly, researchers revealed technological variations that can exist among a number of VFR solutions in the current marketplace primarily due to differences in technology used for visualisation [e.g. AR (camera-based technology), VR (3D simulation)] in addition to differences in body measurement capturing mechanism (Javornik, 2016; Lee and Xu, 2019; Merle et al., 2012). Lee and Xu (2019) identified seven types of VFRs: (1) Full 3D body scanner, (2) 3D avatar, (3) 3D customer model, (4) Photo-accurate 3D customer model, (5) Robotic mannequin, (6) AR fitting room and (7) VR fitting room. Among these seven types of VFRs, AR-based VFRs were found to have been used in the fashion industry to a greater extent compared to the other types of VFRs (Lee and Xu, 2019). Given the unique experience that AR can provide, AR-based services were suggested as having great potential to enhance both utilitarian and hedonic experiences (Caboni and Hagberg, 2019; Javornik et al., 2016; Lee and Xu, 2019). Despite the increasing popularity and great potential to enhance consumer experiences in the fashion industry, the impact of AR features on consumers' experiences was

mainly investigated in the furniture and cosmetics shopping context (Javornik, 2016; Javornik *et al.*, 2016). Therefore, it is important to understand how unique and salient features of AR-based VFRs can contribute to consumers' experiences, which are conducive to shaping consumers' positive attitudes and behaviour.

The theory of interactive media effects

The TIME posits that human responses to media are influenced by technological attributes of the media via consumers' immersive experiences (Sundar *et al.*, 2015). TIME conceptualises technological attributes as underlying affordances, referring to action possibilities presented by perceived properties of a system. Affordance can be psychologically more salient and relevant to users in the interaction, hence, is associated with psychological correlates of users (Sundar *et al.*, 2012, 2015), including perceived control (choice) and sense of agency (self as a source) (Sundar *et al.*, 2015). For example, the mix-and-match features of VFRs can provide more choices to users, giving them the perception of having control over the displayed content. Also, the displayed content can be personalised in terms of users' body dimensions in the case of VFRs, hence, can elicit a sense of agency of self (Javornik, 2016; Merle *et al.*, 2012; Sundar *et al.*, 2015). If the media characteristics are straightforward and accurately signal underlying affordances offered by a system, psychological correlates derived from affordances can subsequently dictate the nature of users' actions towards the system (Sundar *et al.*, 2015). While consumers' strong psychological correlates can exert influences on their responses, it is critical to note the significant mediating role of consumers' immersive experiences (Javornik, 2016). Researchers found that psychological correlates manifested through affordances can bring subsequent cognitive, affective and conative responses with affordance-driven engagement (i.e. absorption, immersion) being the mediating factor (Javornik, 2016; Sundar *et al.*, 2012, 2015). Recognising its ability to disaggregate key media characteristics only in terms of non-message attributes, several researchers used TIME to understand how unique media characteristics solely attributable to systematic features can be translated into consumers' responses (Javornik, 2016; Sundar *et al.*, 2017). For example, Javornik (2016) suggested that psychological correlates elicited from affordances by users can be translated into immersive experiences, which can further lead to subsequent affective, cognitive and conative responses towards AR applications (Apps).

Media characteristics (affordances)

Each medium has a unique technological structure and interactive features, hence, has its own distinct media characteristics (Javornik, 2016). As theorised in TIME, media characteristics are affordances, presented to consumers who psychologically process relevant and salient features and perceive medium properties through their own eyes, which can in turn induce consumers' immersion (Sundar *et al.*, 2015; Yim *et al.*, 2017). In line with TIME, previous studies provide support for the influence of media characteristics on consumers' immersive experiences (Javornik, 2016; Sundar *et al.*, 2015). As AR-based VFRs employ both augmentation features and basic 3D interactive features (Javornik, 2016; Lee and Xu, 2019), perceived interactivity and perceived augmentation were proposed as two important media characteristics of AR-based VFRs that can be translated into immersive experiences, particularly immersive experiences associated with telepresence. In the VFR literature, telepresence, which is characterised by cognitive and sensory immersion, was found to represent key immersive experiences in VFRs due to the interactive nature of virtual try on experiences in VFRs (Chen *et al.*, 2011; Fiore *et al.*, 2005; Klein, 2003).

Interactivity. Interactivity, a salient characteristic of computer-mediated environments, refers to "a psychological state experienced by the user during his or her interaction with the media" from an experiential perspective (Wu, 2006, pp. 91). Interactivity can be classified into

three types, including (1) modality/medium interactivity, referring to “the features and functions that a technology can offer to a user and allow them to take actions and initiate interactions with the medium”, (2) message interactivity, derived from the features that allow “message exchange between different parties” and (3) source interactivity, derived from the ability to “establish the users as the source of communication and the one in control” (Javornik, 2016, pp. 993; Sundar *et al.*, 2015). As postulated in TIME, these are underlying affordances presented to users, which can be associated with psychological correlates such as control, connectedness and responsiveness (Mollen and Willson, 2009; Wu, 2006). Similarly, Mollen and Wilson (2009, pp. 5) defined perceived interactivity as “the degree to which the user perceives that the interaction or communication is two-way, controllable, and responsive to their actions”. For instance, consumers can perceive interactivity as a salient media characteristic of VFRs as they can control the relationship with the stimulus (e.g. changing the angle), modify the stimulus (e.g. changing the product’s feature, mix-and-match) and even can share their images and experiences with others through social media integration (Fiore and Jin, 2003; Fiore *et al.*, 2005; Pachoulakis and Kapetanakis, 2012). Several researchers suggest interactivity as the key media characteristic that can lead to consumers’ immersion, primarily telepresence in the context of VFRs, which create interactive virtual environments (e.g. Fiore *et al.*, 2005; Lee, 2018; Steuer, 1992). Previous researchers found that the characteristics of graphic simulation technology, such as image interactivity, significantly influenced users’ telepresence, which represents flow-like immersive experiences (Fiore *et al.*, 2005; Mollen and Willson, 2009; Steuer, 1992). Thus, we assume:

H1. Consumers’ perceived interactivity of VFRs will have a significant positive influence on their telepresence.

Augmentation. While interactivity is a key media characteristic for all digital technologies, augmentation is a unique and salient characteristic for AR-based applications including AR-based VFRs (Huang and Liao, 2015; Javornik, 2016; Sundar *et al.*, 2015). Unlike VR-based VFRs, AR-based VFRs can provide “mixed reality” by augmenting or superimposing the virtually simulated features, primarily products, on the overlay of users in an augmented physical environment, captured through a camera-based technology embedded in a device (Cho and Schwarz, 2010; Javornik *et al.*, 2016; Yim *et al.*, 2017). It was argued that augmentation is neither the same concept as interactivity nor a sub-dimension of interactivity (Javornik, 2016). Rather, it is a characteristic uniquely attributable to AR-based technology, being a new dimension of media characteristics. Unlike interactivity, which only addresses interaction between the medium and users or among users, augmentation can explain interaction between the medium and the surrounding environment due to its unique features (Caboni and Hagberg, 2019; Javornik, 2016). As an affordance for users, perceived augmentation can be described as the extent to which users can have integrative and realistic experiences through interacting with a virtual environment that is close to the physical environment (Preece *et al.*, 2015). With more para-authentic product experiences offered by augmentation features of AR-based technology, there is greater potential to bring consumers’ immersive experiences compared to VR-based technology (Yim *et al.*, 2017). Similarly, previous studies suggested a significant positive influence of perceived augmentation on immersion such as flow-like experiences (e.g. Javornik, 2016; Javornik *et al.*, 2016). Acknowledging the great potential of AR, Yim *et al.* (2017) also argued that at least as much telepresence as exhibited in VR is expected from AR. Hence, the following hypothesis was developed regarding the influence of media characteristics on consumers’ immersive experiences in AR-based VFRs:

H2. Consumers’ perceived augmentation of VFRs will have a significant positive influence on their telepresence.

Telepresence (immersion)

In the computer-mediated environment, consumers can have flow-like immersive experiences with a medium-induced sense of presence, called telepresence (Steuer, 1992).

According to Steuer (1992, pp. 76), telepresence is described as “the experience of presence in an environment by means of communication medium”. Unlike presence in the immediate physical environment, telepresence measures the extent to which one feels present in the mediated environment. Thus, telepresence has been primarily used in investigating human experiences in mediated-interactive environments such as virtual environments, including VR, IIT and 3D product visualisation technologies (e.g. Fiore *et al.*, 2005; Steuer, 1992; Suh and Chang, 2006). Telepresence can represent users’ cognitive immersion in the mediated environment through the extent to which they feel a sense of presence (Mollen and Wilson, 2009). In previous studies, researchers suggested a positive influence of telepresence on consumers’ attitude and behaviour (Mollen and Wilson, 2009; Suh and Chang, 2006). In the case of graphic simulation technology, consumers showed positive attitude and behavioural intention due to telepresence, signified by enhanced IIT, which can bring consumers’ positive cognitive and affective evaluations in turn (Fiore *et al.*, 2005). Consumers can build positive utilitarian and hedonic aspects of attitudes towards VFRs via telepresence, as a computer-mediated environment that is similar to a real physical environment allows consumers to be better informed about product-related information in addition to enhancing a sense of fantasy, further augmenting their knowledge and shopping enjoyment (Fiore *et al.*, 2005; Song *et al.*, 2007; Suh and Chang, 2006). Similarly, Mollen and Wilson (2009) argued that telepresence can enhance consumers’ instrumental and experiential beliefs, and help consumers by establishing and intensifying a positive attitude towards a product or service. Therefore, we assume:

- H3.* Consumers’ telepresence in VFRs will have a significant positive influence on their attitude (H3a: utilitarian component, H3b: hedonic component) towards VFRs.

In addition, in previous research, telepresence was found to serve as an intermediate variable between media characteristics and consumers’ attitudes and behaviours (Fiore *et al.*, 2005; Mollen and Wilson, 2009; Suh and Chang, 2006). Although media characteristics can potentially serve as symbolic representational cues on the interface if the presented features are psychologically salient enough for users, directly impacting on consumers’ cognitive, affective and behavioural responses (Debbabi *et al.*, 2010; Fiore *et al.*, 2005); previous research also shows that media characteristics are not of themselves enough to generate positive consumers’ responses (Klein, 2003; Mollen and Wilson, 2009; Suh and Chang, 2006). Specifically, research suggested that consumers cannot enhance their psychological responses such as attitudes towards a product or service unless they engage into interactive features and are immersed to the computer-mediated environment (Li *et al.*, 2002; Nah *et al.*, 2011). Therefore, it is vital to understand whether telepresence has a significant mediating effect on consumers’ responses towards AR-based VFRs as immersive experiences associated with telepresence are considered action triggers that improve attitudes and give rise to behavioural intention (Javornik, 2016; Mollen and Wilson, 2009). It is unknown whether media characteristics of AR-based VFRs would create immersive experiences for consumers and mediate the relationship between media characteristics and consumers’ attitudes. Given the relationships among media characteristics, telepresence and consumers’ attitudes identified here, the following hypothesis was proposed:

- H4.* Consumers’ telepresence will mediate the influences of perceived interactivity and perceived augmentation on utilitarian and hedonic components of attitude.

Consumers’ cognitive, affective and conative responses

When faced with innovative technologies such as AR-based VFRs, consumers refer to their experiences as the basis of evaluations, which can guide and inform their behaviour

(Park *et al.*, 2008; Yim *et al.*, 2017). Previous research suggested that multifarious visual cues and interface features presented by advanced interactive technologies can lead consumers to show both cognitive and affective responses, followed by their conative responses (Holbrook and Hirschman, 1982; Park *et al.*, 2008). Specifically, consumers' evaluation of distinct features presented by a new technology and their contingent experiences can be the grounds for their subsequent attitude towards a technology (Chowdhury and Salam, 2015; Sundar, 2004; Van der Heijden and Sangstad, 2003). Consumers' attitudes consist of dual components, including cognitive and affective components (Van der Heijden and Sangstad, 2003). The cognitive component refers to "the utilitarian value that a person associates with the product or service" (Van der Heijden and Sangstad, 2003, pp. 4). This is derived from a way of thinking or feeling in a more cognitively-driven, instrumental and goal-oriented manner. On the other hand, the affective component of attitude represents the hedonic aspects of attitude, derived from a way of thinking, characterised by "affective and sensory orientation of aesthetic or sensual pleasure, fantasy, and fun" (Torrico *et al.*, 2011, pp. 2). These two components do not necessarily come with equal vigour to consumers' minds in their evaluation process (Shiv and Fedorikhin, 1999). When consumers favourably appreciate and show positive cognitive and affective responses towards a given object, they can show conative responses as a consequence (Millar and Tesser, 1986; Park *et al.*, 2008). Prior researchers supported the positive influence of cognitive and affective responses on consumers' conative responses when investigating consumers' responses to online product display such as 3D interactive displays (e.g. Fiore *et al.*, 2005; Kim and Forsythe, 2008; Park *et al.*, 2008). The intention to adopt innovative technologies is one of the key conative responses that predicts the adoption behaviour (Van der Heijden and Sangstad, 2003).

H5. Consumers' attitude (H5a: utilitarian component, H5b: hedonic component) towards VFRs will have a significant positive influence on their intention to adopt VFRs.

Method

Research framework

Based on TIME and the review of literature, a research framework (Figure 1) was developed to guide the investigation of media effects on consumers' adoption towards AR-based VFRs. Structural equation modelling (SEM) from AMOS 22 was used to test the proposed relationships.

Research design

To empirically examine the proposed relationships, survey data collection was conducted. Convenience sampling was used to recruit participants for the survey, and a paper survey was distributed to students in selected classes. The survey purposefully focused on young

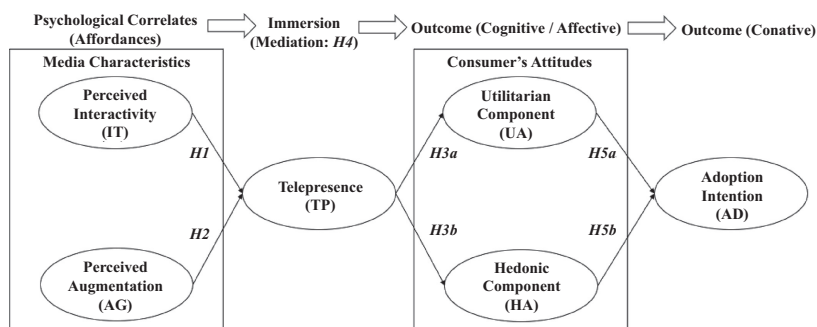


Figure 1. Research framework and hypotheses

consumers who are in their 20s, as they tend to be innovative, digitally-influenced and tech-savvy (Achille *et al.*, 2018). As some participants may not have any prior experience with VFRs, before the survey was distributed, the participants were asked to download an AR-based VFR application, Zeekit, on their mobile phone. Zeekit was selected for this study as it was considered a representative AR-based App for virtual fitting with advanced features (Steele, 2019). It provides personalised fitting process by letting consumers take a picture of themselves in the App and virtually try-on products from a variety of brands (Scarano, 2016). With the use of AR technology, it can render 3D maps over 2D images, accurately detecting users' body dimensions (Scarano, 2016). Also, users can mix-and-match looks, saving the outfits to their virtual closet or sharing their looks with others through a social media connecting tool (Scarano, 2016). Lee and Xu (2019) suggested Zeekit as one of the representative AR-based VFRs currently available in the marketplace. To further validate the use of Zeekit for this study, a pilot test was conducted with 21 students at a US university in November of 2018 via a focus group study. We asked whether Zeekit has the key characteristics of AR-based VFRs (e.g. use of camera-based technology, mixed reality offered, superimpose virtual image of products on augmented self) to determine the appropriateness of using Zeekit as an AR-based VFR. Subjects were recruited for the focus group study via invitations through e-mails and paper flyers distributed in public places on campus. The comments revealed that Zeekit technically could augment the self and physical environment with the virtual additions of a wide variety of products for both men and women. In addition, Zeekit was not well known unlike other VFRs, which use VR-based technology, hence, participants' attitudes towards Zeekit would not be influenced by previously established attitudes. Once Zeekit was selected as the tool to ensure AR-based VFR experiences, the main survey was conducted to examine the proposed relationships.

Data collection procedure

Data were collected from 398 students enrolled at a major university in the Southeast region of the US in February of 2019. Data were collected students enrolled in 13 classes during regular class meeting times. Prior approvals were obtained from IRB and each instructor. During each session, the participants were first asked to download the Zeekit App to their own mobile devices and then take about 10 min to freely explore products and examine their fit. After subjects tried the Zeekit App during the survey session, they were asked to complete the questionnaire in paper format. All students enrolled in the select classes were invited to participate and the participation was voluntary.

Measures

Established scales were used and adapted to the context of VFRs to measure the constructs of this study. Perceived interactivity was measured by nine items from Cyr *et al.* (2009), which consist of control, connectedness and responsiveness. Perceived augmentation was measured by five items from Javornik (2016). Telepresence was measured by five items from Fiore *et al.* (2005). A total of ten items from Voss *et al.* (2003) were used to measure consumers' utilitarian and hedonic attitude components. Three items were adapted from Venkatesh *et al.* (2003) to measure consumers' intention to adopt VFRs. All items were measured on a seven-point Likert scale. The reliabilities of instruments were reported as greater than 0.80 (being 0.81, 0.85, 0.90, 0.88, 0.87 and 0.92, respectively).

Results and discussion

Sample profile

A total of 398 subjects participated in the survey, but only 352 responses were retained for data analysis after 46 incomplete surveys were excluded. As shown in Table 1, the majority of

respondents were female (87.5%). This may be due to a greater number of female students in the classes we visited for data collection, however this proportion reflects the greater number of female consumers in the online apparel market as compared to male consumers (Statista, 2017; Thomas, 2018). About 84.7% of respondents were in their early 20s (i.e. between 20 and 23), followed by the respondents in their ages between 23 and 25 (8.8%). About 92% of respondents were undergraduate students (Freshman: 6.2%; Sophomore: 36.4%; Junior: 31.2%; Senior: 18.2%), followed by 8% of graduate students. About 36.4% of the respondents indicated that they shop for apparel products online once a month, followed by the respondents who shopped more than twice a month (32.9%), and twice a month (21.6%). The majority of respondents (86.3%) indicated that they have experienced problems with buying apparel products online in terms of fit. About 8.5% of the respondents indicated that they had experienced VFRs in the past. Among those who had experienced VFRs in the past, one-third expressed that they were not satisfied with the current VFRs in the marketplace.

Demographic factors	Count (N)	Percentage (%)
<i>Gender</i>		
Female	308	87.5
Male	44	12.5
<i>Age</i>		
20–22	298	84.7
23–25	31	8.8
25–27	11	3.1
28–29	12	3.4
<i>Education</i>		
Freshman	22	6.2
Sophomore	128	36.4
Junior	110	31.2
Senior	64	18.2
Graduate student	28	8.0
<i>Shopping frequency</i>		
Once a month	128	36.4
Twice a month	76	21.6
More than twice a month	116	32.9
None	32	9.1
<i>Fit problem in online</i>		
Definitely yes	148	42.0
Moderately yes	74	21.0
Slightly yes	82	23.3
Neutral	16	4.5
Slightly no	22	6.3
Moderately no	2	0.6
Definitely no	8	2.3
<i>VFR experience</i>		
Yes	30	8.5
No	322	91.5
<i>VFR satisfaction (If yes)</i>		
Yes	8	2.3
Moderately	12	3.4
No	10	2.8
<i>Total</i>	<i>352</i>	<i>100</i>

Table 1.
Sample profile

Measurement model

The two-step approach of SEM suggested by [Anderson and Gerbing \(1988\)](#) was adopted to test [H1-H5](#) regarding the influence of media characteristics on consumers' telepresence, attitudes and adoption intention of VFRs. A confirmatory factor analysis (CFA) was first conducted to assess the model fit, and the results are illustrated in [Table 2](#). Per the model fit criteria proposed by [Hu and Bentler \(1999\)](#), the overall fit of the measurement model was deemed satisfactory: $\chi^2 = 1539.653$ ($df = 449, p = 0.000$), SRMR = 0.043 (< 0.08), TLI = 0.913 (> 0.9), IFI = 0.921 (> 0.9), CFI = 0.921 (> 0.9) and RMSEA = 0.063 (< 0.08). Significant t -values of factor loadings (λ) indicated good convergent validity ([Table 2](#)). Also, Cronbach's alpha values of all constructs were greater than 0.70, composite reliability values were greater than 0.70 and average variance extracted values (AVE) were greater than 0.50. The results indicated that the measurement model was internally consistent, implying convergent validity. Correlations among the seven latent variables were less than 0.90, indicating that a multi-collinearity problem was non-existent. All AVEs were greater than the squared correlations (R^2), achieving discriminant validity ([Hu and Bentler, 1999](#)) ([Table 3](#)).

Structural model

Single group SEM was used to test [H1-H5](#). The overall fit indices showed a satisfactory fit to the data: $\chi^2 = 1660.655$ ($df = 453, p = 0.000$), SRMR = 0.059, TLI = 0.904, IFI = 0.913, CFI = 0.913, RMSEA = 0.067. The maximum likelihood estimation was used to estimate the path coefficients. The SEM results are depicted in [Figure 2](#).

Regarding the influence of perceived interactivity and augmentation on telepresence, both media characteristics showed a positive influence on telepresence ($\beta = 0.245, p < 0.001$; $\beta = 0.455, p < 0.001$ respectively), supporting hypotheses [H1](#) and [H2](#). With respect to the influence of telepresence on consumers' attitude, telepresence had a positive influence on both utilitarian ($\beta = 0.462, p < 0.001$) and hedonic ($\beta = 0.189, p < 0.001$) components of attitude towards AR-based VFRs, supporting [H3a](#) and [H3b](#). In terms of the influence of utilitarian and hedonic components of attitude on the intention to adopt AR-based VFRs, both components showed a significant positive influence on the adoption intention ($\beta = 0.445, p < 0.001$; $\beta = 0.439, p < 0.001$ respectively), supporting [H5a](#) and [H5b](#).

Mediating effect: telepresence

The mediating effect of telepresence ([H4](#)) was analysed using the bootstrapping method. First, the influence of perceived interactivity and augmentation on consumers' attitude was examined by controlling the mediator. Perceived interactivity ($\beta = 0.400, p < 0.001$; $\beta = 0.364, p < 0.001$ respectively) and augmentation ($\beta = 0.406, p < 0.001$; $\beta = 0.249, p < 0.001$ respectively) showed a significant influence on utilitarian and hedonic components of consumers' attitude. Therefore, the significance of the influence of media characteristics on attitude was confirmed. However, when telepresence was introduced in the model, there was no direct effect of perceived augmentation on consumers' hedonic attitude ($\beta = 0.105, n.s.$), indicating a complete mediation through telepresence ([Table 4](#)). On the other hand, there was a significant direct effect of perceived interactivity and augmentation on consumers' utilitarian attitude ($\beta = 0.285, p < 0.001$; $\beta = 0.149, p < 0.01$ respectively) and direct effect of perceived interactivity on hedonic attitude ($\beta = 0.361, p < 0.001$), indicating partial mediation through telepresence. Specifically, the indirect effects of perceived interactivity on utilitarian and hedonic components of attitude were about 0.195 ($p < 0.01$) and 0.326 ($p < 0.01$) respectively. Also, the indirect effects of perceived augmentation on utilitarian and hedonic components of attitude were about 0.326 ($p < 0.01$) and 0.246 ($p < 0.01$). Thus, the indirect influence of perceived interactivity and augmentation on consumers' attitude through the mediation of telepresence can be found, supporting [H4](#).

					Consumers' adoption of AR-based VFRs
Item	λ factor loading	C.R.	α	AVE	
<i>Perceived Interactivity (IT)</i>					
I was in control over the information display format, conditions when using the Zeekit app	0.782***	0.955	0.936	0.704	
I was in control over the content of the Zeekit app that I wanted to see	0.795***				
Customers of the Zeekit app can interact with other members of the Zeekit community	0.819***				
Customers can share common interests with other members of the customer community using the Zeekit app	0.794***				
When I interacted with the Zeekit app, the information shown was relevant	0.839***				
When I interacted with the Zeekit app, the information shown was appropriate	0.850***				
When I interacted with the Zeekit app, the information shown met my expectations	0.866***				
When I interacted with the Zeekit app, the information shown was suitable	0.915***				
When I interacted with the Zeekit app, the information shown was useful	0.881***				
<i>Perceived Augmentation (AG)</i>					
The Zeekit app can add products to my body virtually	0.841***	0.950	0.949	0.791	
The way the products were virtually placed on my body seemed real	0.916***				
The virtual products seemed to be part of my virtual body	0.900***				
The level of reality seemed high on the App	0.903***				
The products seemed to exist in real time	0.886***				
<i>Telepresence (TP)</i>					
The Zeekit app let me easily visualise what the actual garment is like	0.902***	0.965	0.965	0.848	
The Zeekit app gave me as much sensory information about the products as I would experience in a store	0.922***				
The Zeekit app created a product experience similar to the one I'd have when shopping in a store	0.934***				
The Zeekit app allowed me to interact with the products as I would in a store	0.932***				
The Zeekit app provided accurate sensory information about the products	0.913***				
<i>Utilitarian Component of Attitude (UA)</i>					
Using the Zeekit app for apparel shopping would be effective	0.935***	0.956	0.956	0.813	
Using the Zeekit app for apparel shopping would be helpful	0.908***				
Using the Zeekit app for apparel shopping would be functional	0.908***				
Using the Zeekit app for apparel shopping would be necessary	0.873***				
Using the Zeekit app for apparel shopping would be practical	0.884***				
<i>Hedonic Component of Attitude (HA)</i>					
Using the Zeekit app for apparel shopping would be fun	0.912***	0.972	0.972	0.874	
Using the Zeekit app for apparel shopping would be exciting	0.948***				
Using the Zeekit app for apparel shopping would be delightful	0.944***				
Using the Zeekit app for apparel shopping would be thrilling	0.933***				
Using the Zeekit app for apparel shopping would be enjoyable	0.938***				
<i>Adoption Intention (AD)</i>					
I would like to use Zeekit app for apparel shopping in the future	0.953***	0.957	0.957	0.880	
I would like to use other VFR apps like Zeekit for apparel shopping in the future	0.936***				
I would recommend my friends to use VFR apps like Zeekit for apparel shopping	0.925***				
Note(s). ***. $p < 0.001$					

Table 2.
Confirmatory factor analysis results

Table 2.
Confirmatory factor analysis results

Table 3.
AVE values and
squared correlation of
each latent variable

Conclusions, implications and limitations

The current empirical study examined the influences of media characteristics of AR-based VFRs on telepresence, consumers' attitudes and adoption intention. Whereas previous research (e.g. [Fiore et al., 2005](#); [Lee et al., 2010](#)) has examined the effect of technology attributes on consumers' responses towards VR-based VFRs, the current study is the first to investigate the influences of media characteristics, uniquely attributable to AR-based VFRs, on consumers' cognitive, affective and conative responses from the consumer perspective. Academically, the findings add to empirical support for the relationships proposed in TIME ([Sundar et al., 2015](#)) as well as extending it in an emergent AR-based VFR context, confirming the influences of media characteristics on bringing immersive experiences, enhancing consumers' attitudes and increasing adoption intention towards AR-based VFRs. In addition,

	IT	AG	TP	UA	HA	AD
IT	<i>0.704</i>					
AG	0.382	<i>0.791</i>				
TP	0.278	0.368	<i>0.848</i>			
UA	0.377	0.366	0.494	<i>0.813</i>		
HA	0.300	0.240	0.216	0.539	<i>0.874</i>	
AD	0.315	0.279	0.242	0.527	0.517	<i>0.880</i>

Note(s): The average variance extracted (AVE) are in boldface. The squared correlations (R2) of all constructs are on the off-diagonal

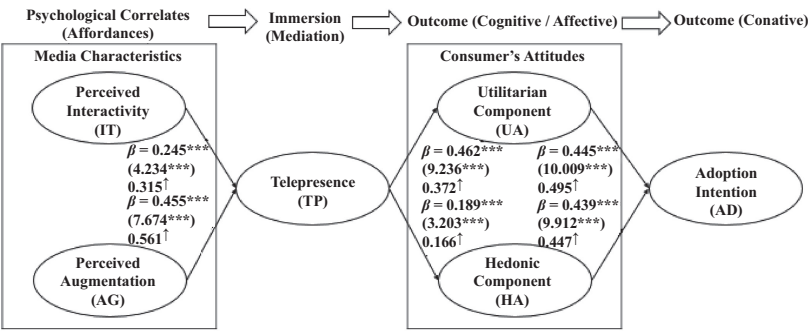


Figure 2.
Structural
equation model

Note(s):
1) t -values are given in parentheses
2) *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$
3) $\dagger$: Estimates for the paths

Table 4.
Mediation effect
analysis results

Paths	Direct effect without mediator	Direct effect with mediator	Indirect effect
IT→TP→UA	0.400 ^{***}	0.285 ^{***}	0.195 ^{**} (Partial mediation)
AG→TP→UA	0.406 ^{***}	0.149 ^{**}	0.147 ^{**} (Partial mediation)
IT→TP→HA	0.364 ^{***}	0.361 ^{***}	0.326 ^{**} (Partial mediation)
AT→TP→HA	0.249 ^{***}	0.105	0.246 ^{**} (Full mediation)

Note(s). **: $p < 0.01$, ***: $p < 0.001$

by validating the mediating role of telepresence, results provide evidence that VFR experiences created by telepresence in AR-based VFRs can be perceived as real and immersive by consumers with advanced media characteristics, enough to stimulate their cognitive, affective and conative responses. Practical implications can be also drawn from this study for fashion firms with respect to managerial decisions regarding the adoption of AR-based VFRs. In addition, fashion firms can effectively formulate optimised marketing strategies to attract fashion consumers to use AR-based VFRs, by addressing what drives adoption. Furthermore, the results provide a guideline for VFR service providers about the way to effectively render interactive and augmented features of AR-based VFRs.

First, the result underscores the superior nature of AR-based VFRs in their ability to induce immersive experiences associated with telepresence. With combined unique AR and interactive features, consumers could feel present in AR-based VFRs, demonstrating the sense of being in the mediated environment. Interestingly, not only did 3D interactive features typically adopted across VFRs in general induce telepresence, but also augmentation features, which are unique to AR-based VFRs, contributed to immersive experiences. As the sensation of telepresence, derived from powerful and superior media characteristics, can lead to cognitive, affective and conative responses that are normally set off by the direct experience (Debbabi *et al.*, 2010), it is recommended for fashion firms to utilise AR-based VFRs instead of VR-based VFRs to increase the chances of consumers being immersed in the mediated environment. Also, VFR service providers should consider enhancing both interactive features (e.g. interactive communication features) and augmentation features (e.g. motion-capturing features) by incorporating AR technologies to their VFR solutions instead of merely focusing on improving the congruity between an avatar and consumer's figure via enhanced 3D simulation techniques.

In terms of consumers' responses to salient media characteristics of AR-based VFRs, telepresence mediated the effects of media characteristics on consumers' cognitive and affective responses towards AR-based VFRs. Advanced features of AR-based VFRs may induce utilitarian and hedonic attitudes as well as adoption intention through the mediation of telepresence as it can place users into more realistic environment with better immersing consumers (Yim *et al.*, 2017). Specifically, telepresence fully mediated only the relationship between perceived augmentation and the hedonic component of consumers' attitude towards the AR-based VFR application. This implies that augmentation of the self and the surrounding space allows consumers to build hedonic attitude towards AR-based VFRs, triggering consumers' enjoyment of AR-based VFRs, because consumers immerse to the mediated environment. Augmentation features can provide more integrative and realistic experiences by permitting the overlap of virtual elements in the physical world (Caboni and Hagberg, 2019). Thus, consumers can regard AR-based VFRs as hedonically pleasing, feeling the sense of being present and further facilitating their fantasy that entails the pleasurable mental imagery involving product use (Caboni and Hagberg, 2019; Song *et al.*, 2007). That is, consumers interacting with AR-based VFRs view the mediated AR environment as a "space of fantasy" when their perceptions in the mediated environment can dominate those in non-mediated environment via AR features (Caboni and Hagberg, 2019; Song *et al.*, 2007). As telepresence is a key mediating factor that enables consumers to form hedonic attitudes, VFR service providers should further enhance augmentation features of AR-based VFRs, for example, with motion-capturing features that allow real-time interaction, to induce telepresence enough to facilitate consumers' experiential pleasure. In addition, as the mere presence of augmentation features does not directly contribute to the formation of consumers' hedonic attitudes, fashion firms should adopt an AR-based VFR that has the best technical ability to immerse consumers to an AR space via enhanced AR features that stimulate clearer mental imagery and fantasy and allow smooth interactions between virtual elements and augmented real elements.

In comparison, interestingly, telepresence only exerted a partial mediation on the influence of augmentation on utilitarian attitude. Augmentation features provide consumers the opportunity to try on virtual garments in a more practical and realistic atmosphere. Consumers can have more para-authentic product experiences as AR features allow them to see virtual garments in a realistic setting (Chen *et al.*, 2011; Yim *et al.*, 2017). As AR features allow consumers to have more complete information by incorporating not only virtual components but also physical and imagery components, consumers can regard AR-based VFRs as providing utilitarian benefits during the online shopping process (Chen *et al.*, 2011). With AR features perceived as the most salient and innovative media characteristics of AR-based VFRs, not only a sensation of telepresence but also their mere perceptions of augmentation led consumers to build utilitarian attitudes towards AR-based VFRs. These findings are in line with TIME, which argues that users' psychological process in the interactive media can come from media characteristics through two routes, action and cue. While users' immersive experiences primarily come from the action route where psychological correlates, derived from affordances, lead to immersive experiences; users can directly show their cognitive and affective responses even with a mere presence of rich affordances through the cue route (Sundar *et al.*, 2015). In this respect, fashion firms should promote and adopt AR-based VFRs to enhance consumers' utilitarian experiences even if AR features are not advanced enough to provide immersive experiences at the current stage, as consumers would build their utilitarian attitude not only because of telepresence but also perceived augmentation. It is desirable to keep showcasing AR-based VFRs in retail stores as consumers would appreciate AR features that provide utilitarian benefits. In addition, it is critical for VFR providers to create an environment that enhances consumers' telepresence in order to streamline the effect of perceived augmentation on consumers' utilitarian attitude.

Similarly, telepresence was found to have only a partial mediating role in the relationship between perceived interactivity and consumers' utilitarian and hedonic attitudes. As compared to AR features, which are more likely to be eye-catching features of AR-based VFRs, consumers can acknowledge the role of the interactive features in their experience (Debbabi *et al.*, 2010). Thus, it is possible that consumers may be not completely feel telepresence in the computer-mediated environment simply due to perceived interactivity (Debbabi *et al.*, 2010; Javornik, 2016). In other words, consumers can form attitudes on their VFR experiences based on telepresence but also on their perceptions of interactivity because consumers can understand how interactivity contributes to their utilitarian and hedonic experiences. This finding supports previous research in that interactivity could be perceived as useful and fun for consumers as they are comparatively accustomed to interactive features, a commonly observed media characteristic for digital technologies (Javornik, 2016; Mollen and Willson, 2009). In this respect, VFR service providers should incorporate advanced interactive features to enhance consumers' telepresence, which can further strengthen consumers' attitudes. Also, fashion firms should adopt AR-based VFRs that have key interactive features, which consumers would typically expect from virtual experiences, to help them build utilitarian and hedonic attitudes towards AR-based VFRs. In addition, fashion firms should emphasise the availability of interactive features in their marketing as consumers would form their utilitarian and hedonic attitudes anyway in their evaluation of VFR experience.

Lastly, both utilitarian and hedonic components of attitudes built upon consumers' evaluations of distinct features and contingent experiences drove consumers' adoption of AR-based VFRs. In other words, consumers' decision to adopt AR-based VFRs resulted from the combined evaluations of cognitive and affective aspects. Therefore, to increase consumers' adoption of AR-based VFRs, fashion firms should promote VFRs as not only a functionally useful option but also as an experientially fun option in their marketing considering the integrative nature of consumers' evaluative process. Also, VFR service providers should find

ways to further enhance both AR and interactive features, which serve as the catalyst to bring consumers' cognitive, affective and conative responses.

Despite the meaningful implications derived from this study, the findings reflect only limited aspects of user experiences with AR-based VFRs. One limitation is the use of student samples as survey respondents. The sample in the present study was predominantly female and aged in their 20s. Therefore, caution needs to be placed when generalising the findings of this study. Future research may replicate the current study using a larger sample with more diverse demographics to obtain more generalisable results. We also acknowledge that the potential limitation lies in the controllability of respondents' experiences with the Zeekit App. Some variations in respondents' mobile device specifications (e.g. varying time taken for the installation and processing) were observed during the survey due to different levels of privacy protection in the device settings. Nonetheless, this study can provide valuable insights into the adoption of this relatively unexplored technology in online fashion shopping. Future studies could consider offering devices with privacy protective measures to respondents to control the variations in the technical specifications.

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